Amazon India Sales Dataset Analysis

Data Analysis Report (31 March – 29 June) /2022

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Executive Summary

This report analyzes Amazon India sales data from March 31 to June 29, 2022. The dataset contains 23 columns and 128,976 rows. The analysis examines sales, orders, returns, and revenue, with a focus on fulfillment methods, product categories, sizes, shipping methods, and buyer types.

The analysis revealed a gradual decline in total sales and received orders during the period. It also showed clear differences in sales and order volumes between Amazon and Merchant fulfillment methods. Additional findings on buyer types, product categories, sizes, shipping methods, and sales forecasts are presented in the following sections.

Business Objective

To analyze sales based on orders, returns, and order-related revenue, while examining their relationship with fulfillment processes, product categories, sizes, and buyer types across the available months, and to identify unexpected patterns and insights in the data.

Data Overview

This report analyzes data for the period from March 31 to June 29, 2022. The original dataset comprises 23 columns and 128,976 rows. During data preparation, a conditional column was added to classify orders, resulting in 24 columns in the analyzed dataset. Some original columns were excluded from the analysis because they were not relevant to the selected analytical objectives. Key columns used in the analysis include Order Status, Buyer Type, Fulfillment, Ship-Service-Level, Category, Size, and Amount. Column names and categories were also adjusted for clarity.

Data Preparation

- Data Collection: The raw Amazon sales dataset was obtained from Kaggle and used as the primary data source for this analysis.

- Data Cleaning: Removing duplicate records, empty rows, and other data inconsistencies identified during the preparation process.
- Standardization: Harmonizing data formats, including standardizing date formats and removing unnecessary timestamps, to ensure consistency across records. Additionally, an "**Order Status**" column was created using the **IF function** to categorize orders based on shipping status and delivery agent status.
- Data Transformation: Renaming the “B2B” column to “Buyer Type” and updating the categories from “Yes/No” to “Retail/Customers” and “Corporate/Business” for greater clarity. A conditional column was also added to classify orders based on shipping status and courier status.
- Validation: Verifying the prepared data to ensure its quality, consistency, and suitability for the intended analysis.

Analytical Approach

The analysis was conducted using Excel and Power Query. After preparing and validating the data, PivotTable analysis was used to address the established research questions, along with basic descriptive statistics.

The analysis examined sales and order patterns across fulfillment methods, buyer types, order sizes, product categories, and shipping methods. To support comparisons, the average sales value per order was calculated across these dimensions. Correlation analysis used the three most recent months (91 days, April–June). The correlation between received orders and sales was **0.999**, while the correlation between Expedited and Standard orders was **0.988**, indicating very strong positive relationships.

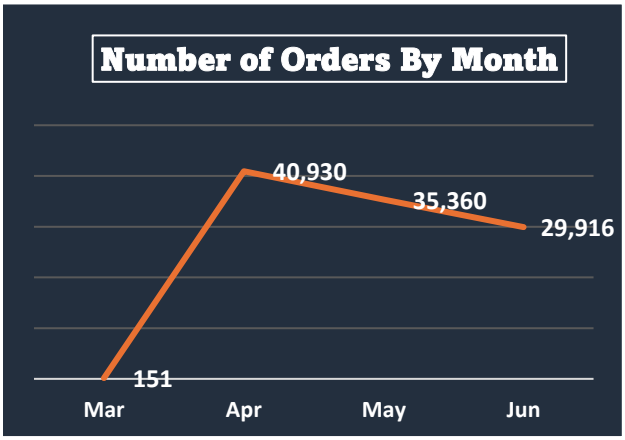
Finally, the Excel Forecast Sheet feature was used to project future trends in received orders and overall sales, including a specific forecast for the Business segment.

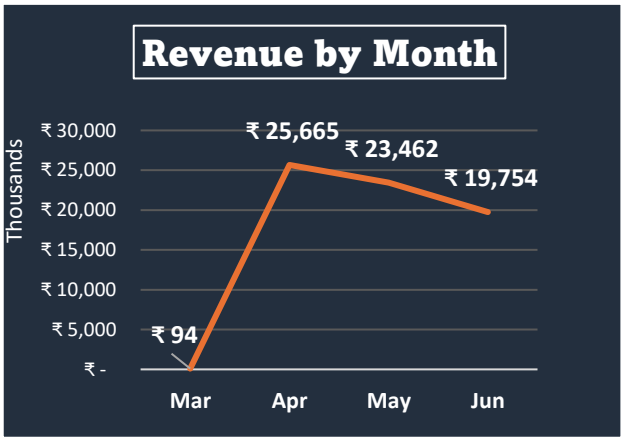
Key Findings

Metric	Result
Top category by sales value and received orders	**Set — ₹34,374,975**
Top size by sales value and received orders	**Medium — ₹12,078,358**
Top buyer type by sales and orders	**Retail / Customers — ₹68,429,433; 105,578 orders**
Average order value — Corporate / Business	**₹700.4**
Average order value — Retail / Customers	**₹648.1**
Top fulfillment method by sales and received orders	**Amazon — ₹50,324,255; 77,596 orders**
Business average order value — Amazon	**₹685.6**
Business average order value — Merchant	**₹732.0**

1. Sales and Order Trends

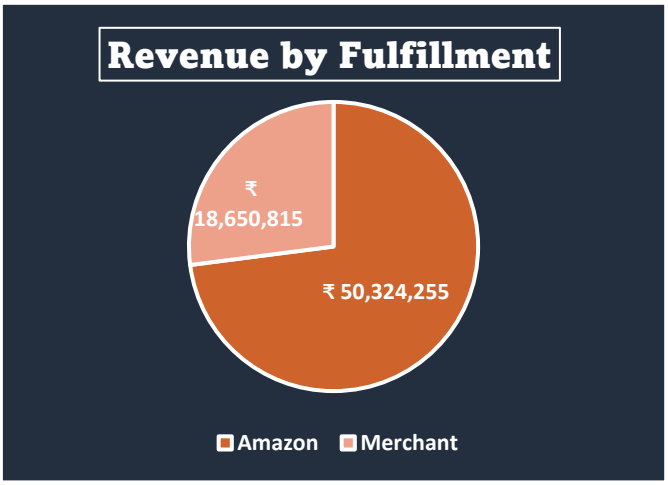
Data from April through June reveals a clear decline in both total sales and order volume. The total number of orders decreased month-over-month, while the proportion of unreceived orders relative to total orders increased during the same period.

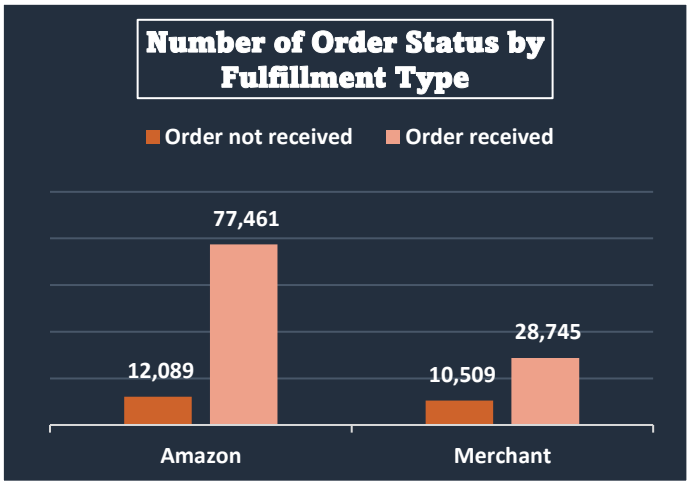




2. Fulfillment Performance: Amazon vs. Merchant

Amazon recorded higher total sales and a higher number of received orders than Merchant. Amazon generated **₹50,324,255** in received sales from **77,596 received orders**. Overall, the average sales value per order was relatively similar between the two fulfillment methods. However, within the **Corporate/Business segment specifically**, Merchant recorded a higher average order value than Amazon.

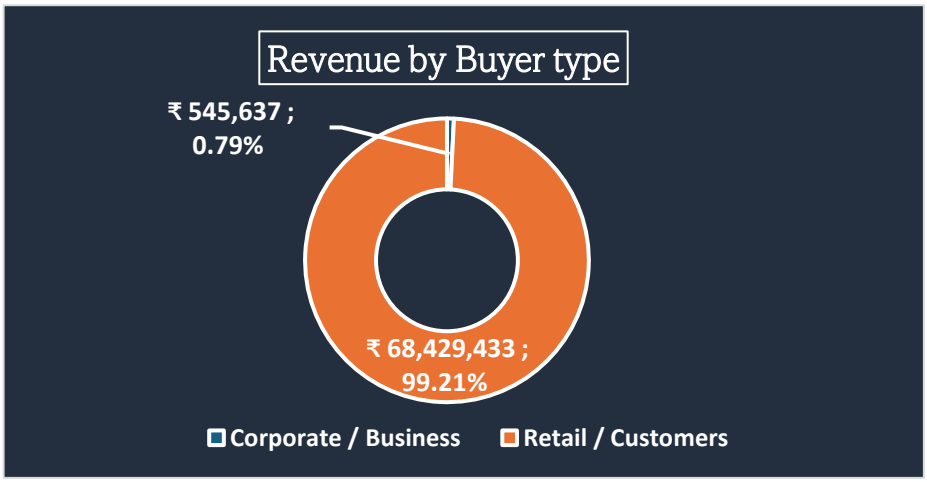




3. Buyer Type

Retail/Customers generated the highest total sales and order volume, accounting for ₹68,429,433 (99.21%) of total sales and 105,578 orders .

In comparison, Corporate/Business generated **₹545,637 (0.79%)** from **779 orders**. However, Corporate/Business orders had a higher average order value of **₹700.4**, compared with **₹648.1** for Retail/Customers. This indicates that Corporate/Business order s have a higher average transaction value despite representing a much smaller share of total sales and orders.



4. Categories and Sizes

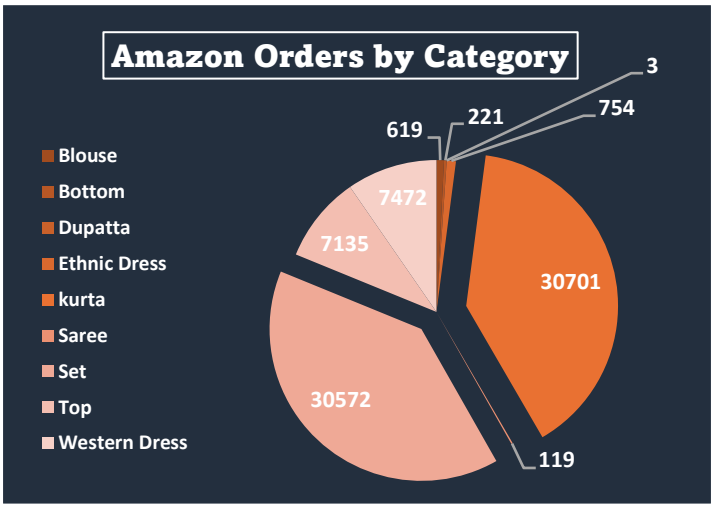
The **Set** category recorded the highest total sales and the highest number of received orders, with total sales of **₹34,374,975**.

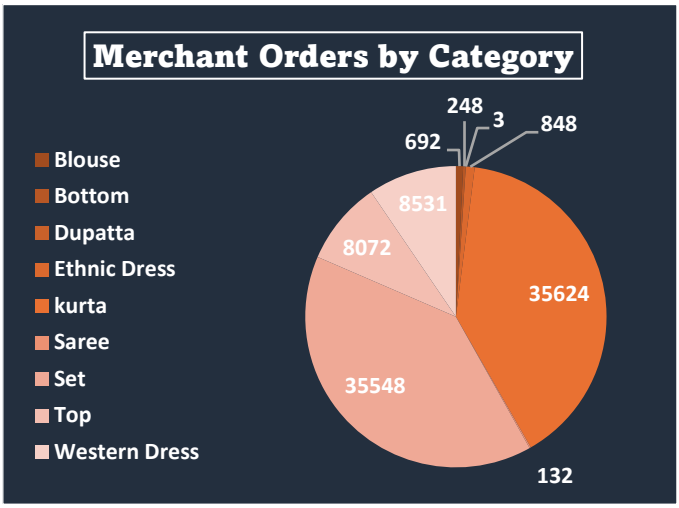
Among sizes, Medium was the highest overall in sales and order volume, followed by Large. Within the Corporate/Business segment, Large had the highest order volume, while Medium had the highest sales, partly due to its higher average order value (652 vs. 636).

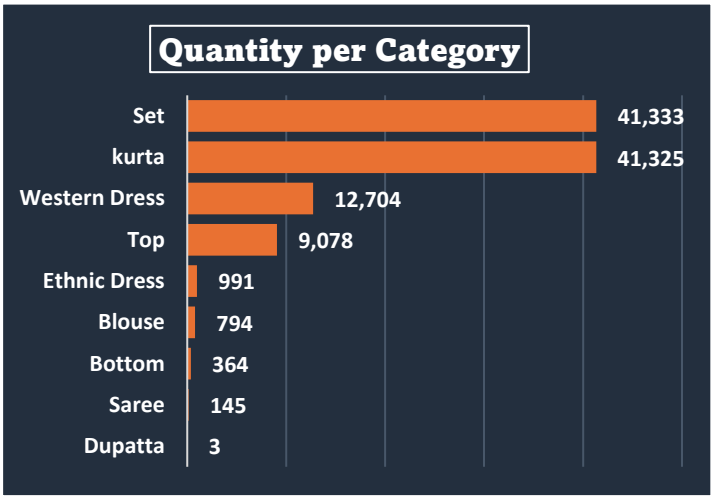
Within **Amazon fulfillment**, **Kurta** had the highest quantity and order volume, while **Set** generated higher total sales. This difference indicates that sales performance is influenced not only by the number of units sold but also by the value of individual orders.

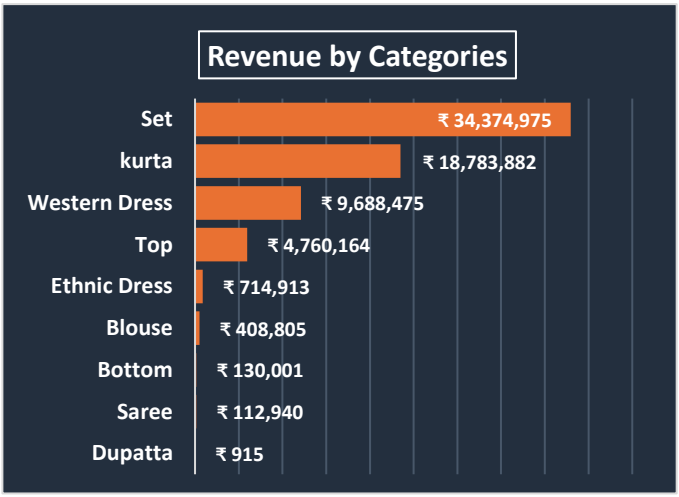
Within the **Corporate/Business segment**, **Large** had the highest order volume, while **Medium** generated the highest total sales. The higher average order value of **Medium** helps explain why it generated more sales despite having fewer orders than **Large**.

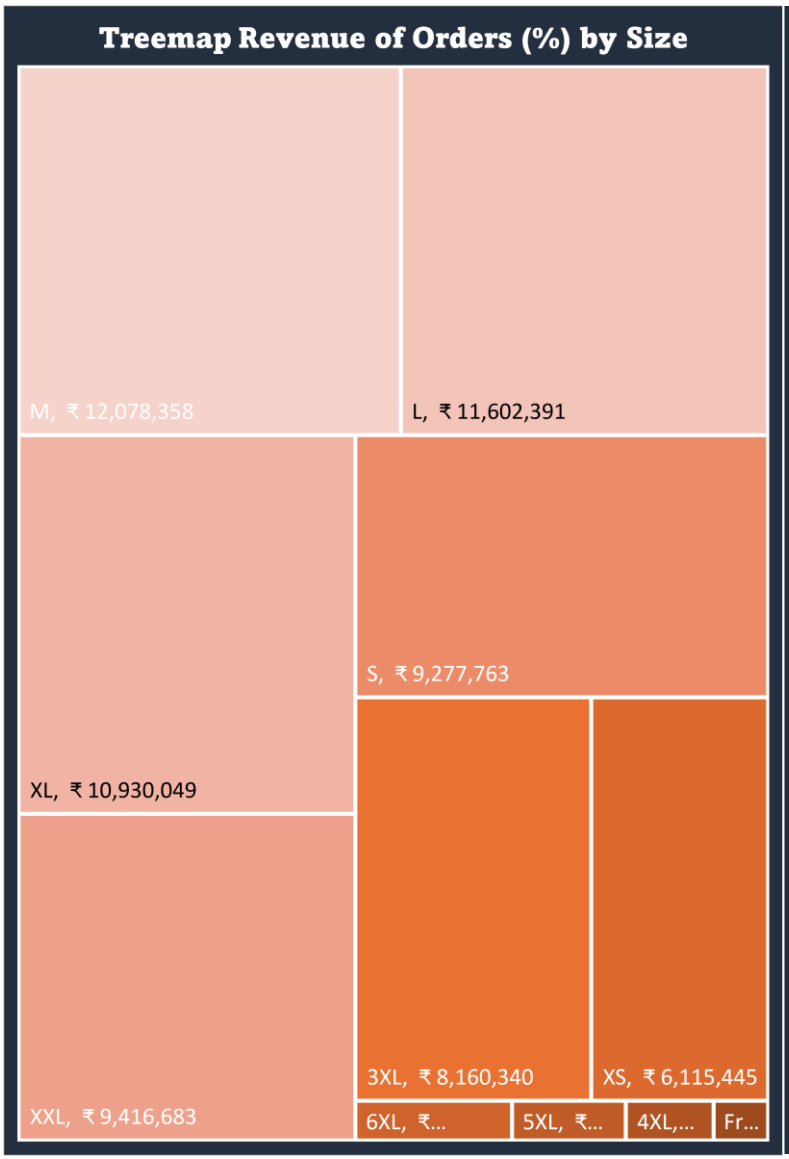
Note: The following charts are unfiltered; you can filter them using the Excel file.

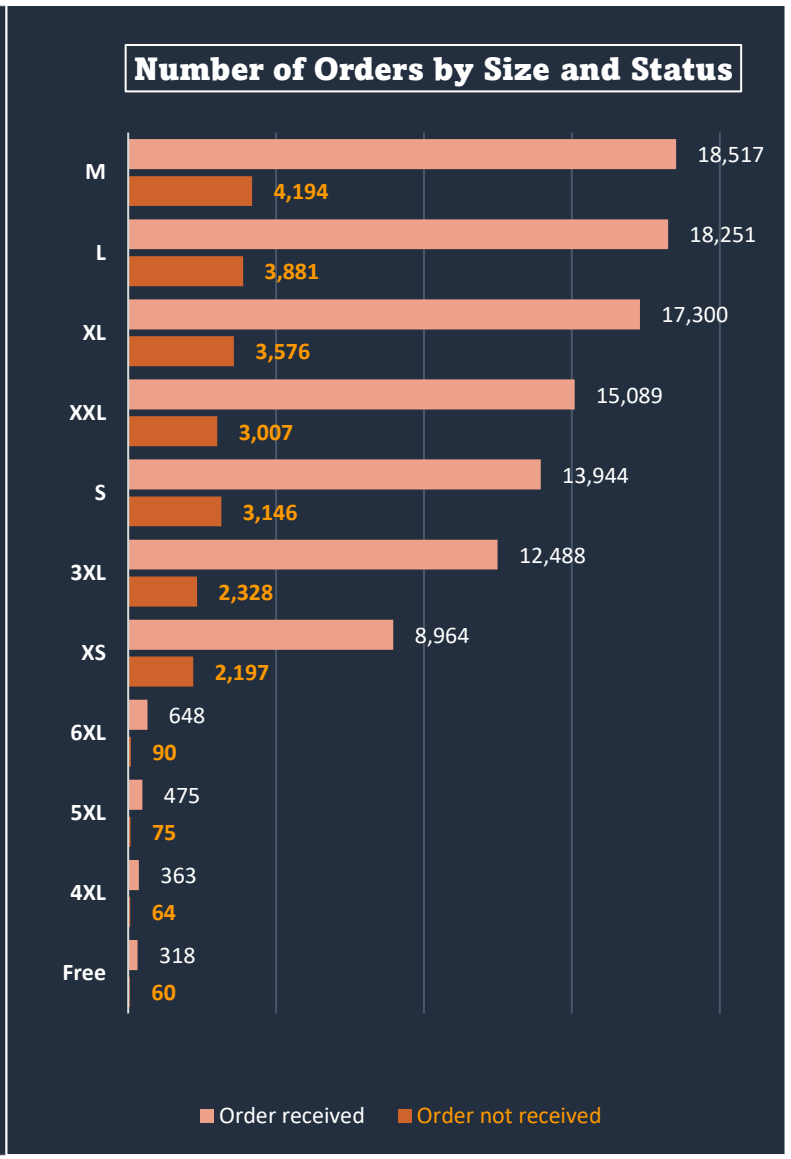










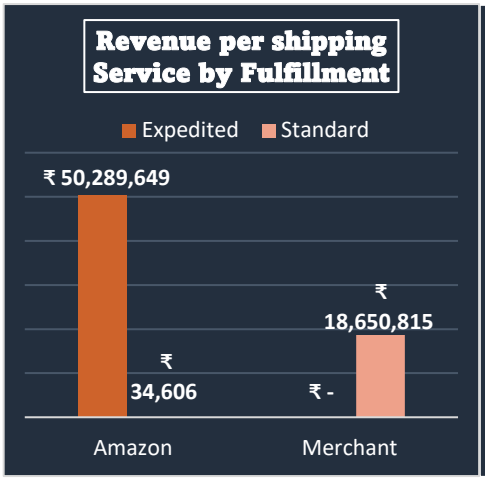


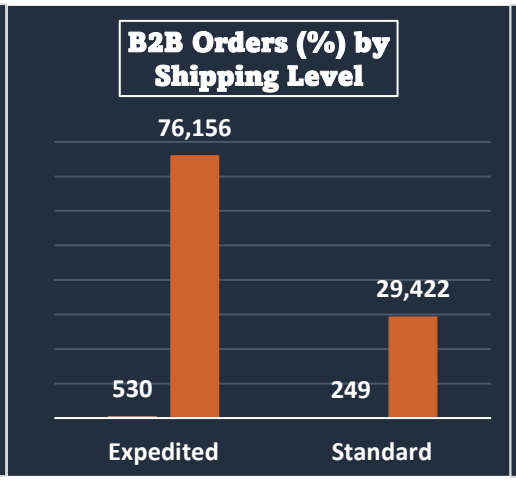
5. Shipping Patterns

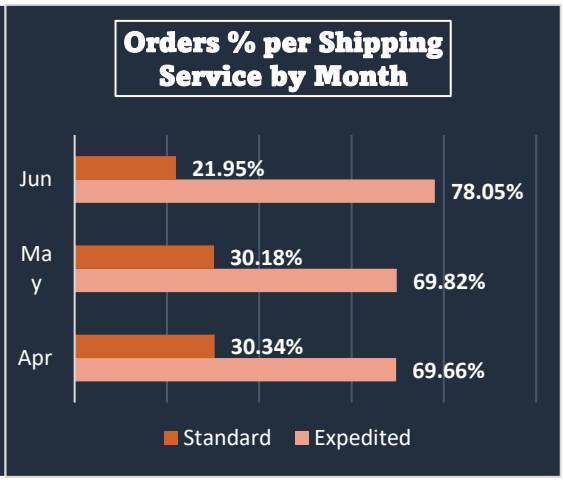
The analysis showed that received orders were more concentrated in **Expedited** shipping than in Standard shipping

Amazon's orders were heavily concentrated in Expedited shipping, while Merchant orders relied predominantly on Standard shipping. Across the analyzed months, the share of Expedited shipping increased while the share of Standard shipping decreased.

The analysis also showed a difference in shipping preferences by buyer type.
Corporate/Business orders relied more heavily on Expedited shipping than on Standard shipping, indicating a greater use of faster shipping within this segment.

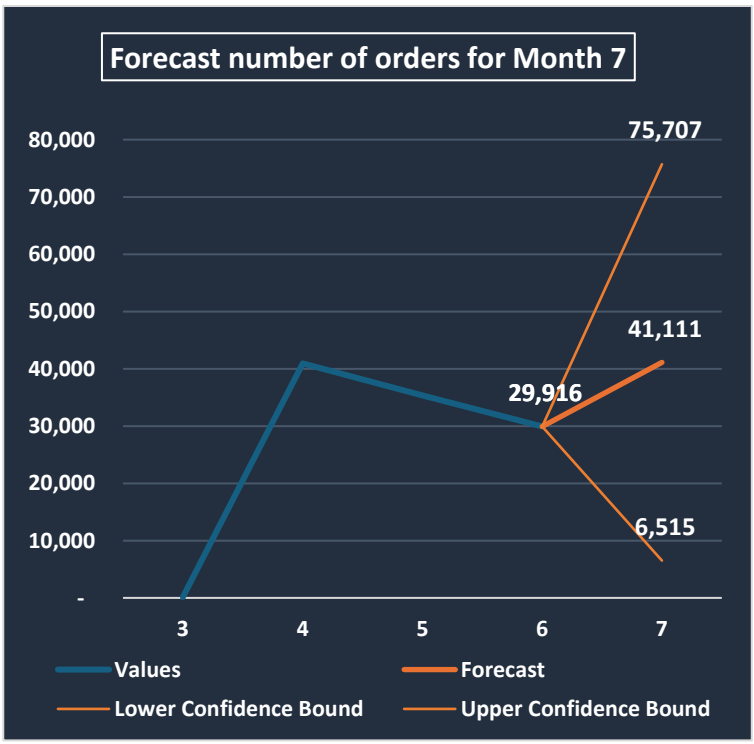


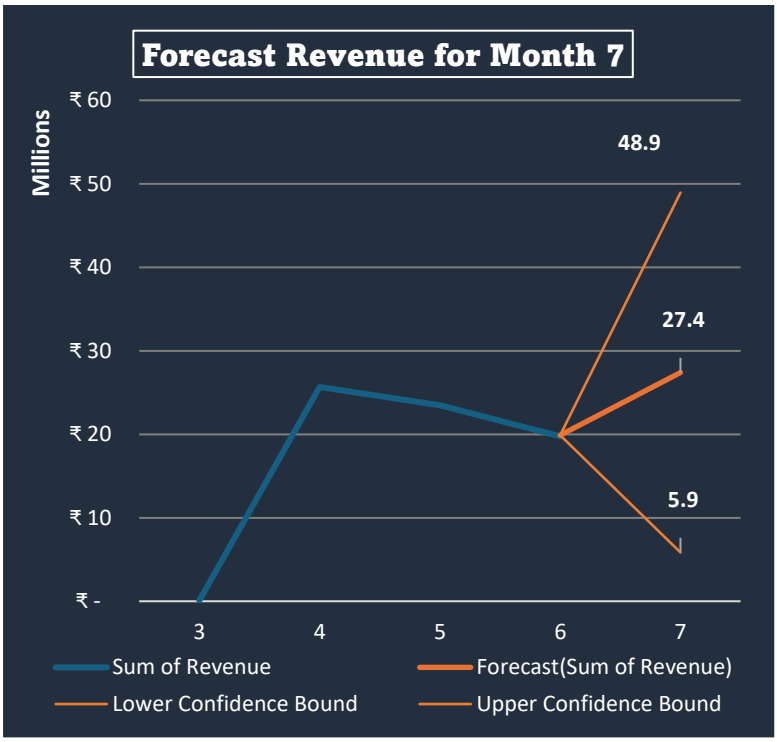




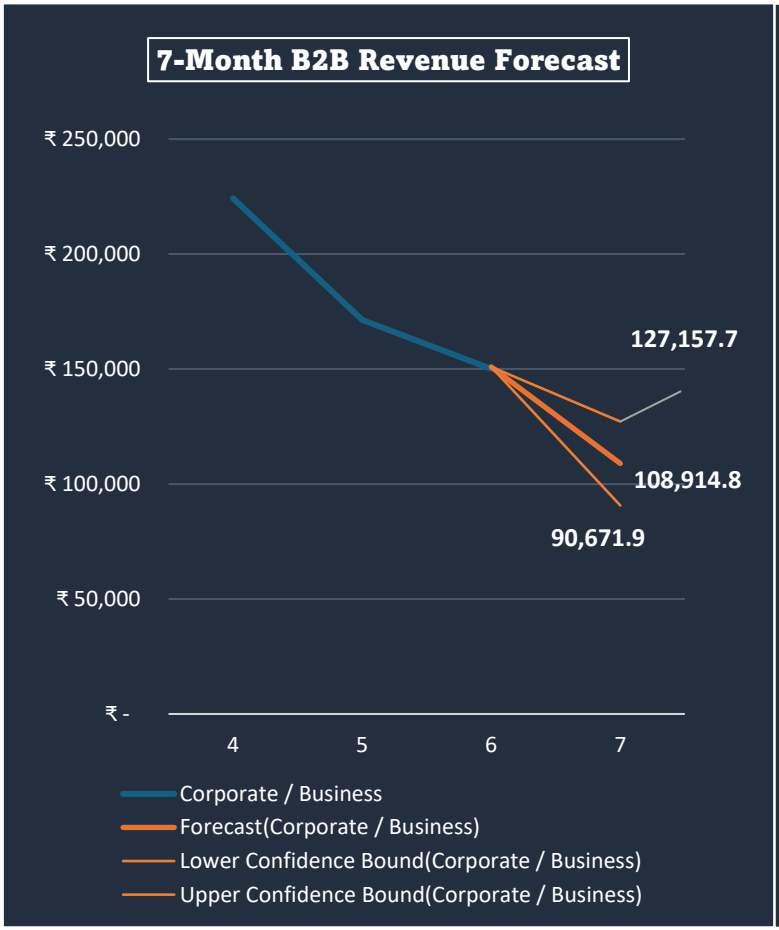
6. Forecast

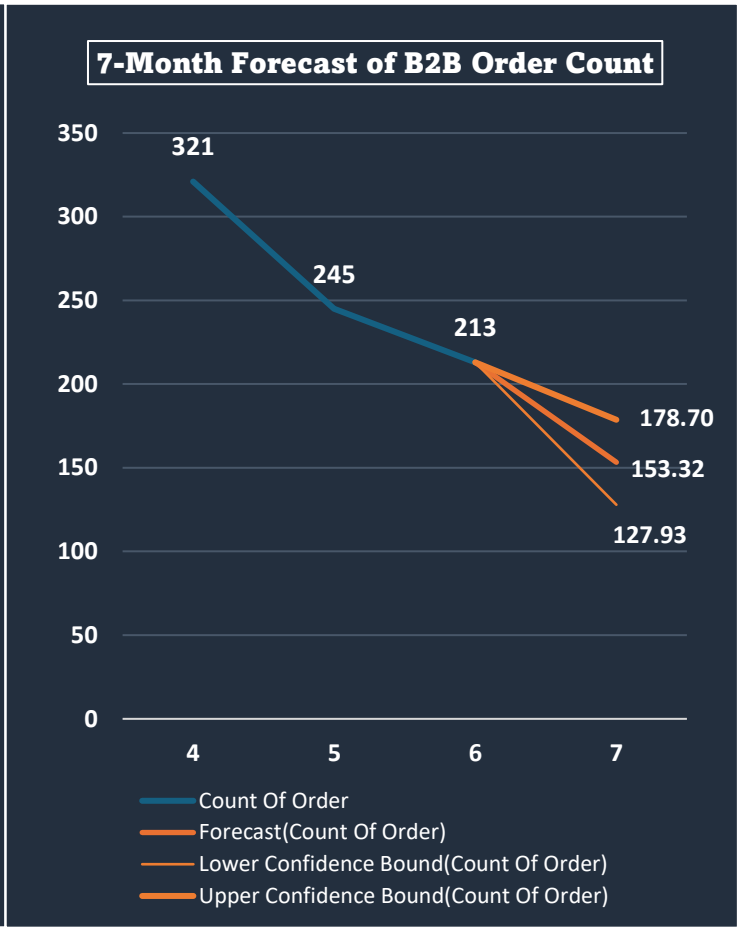
The overall forecast indicates fluctuations in both sales and order volume. The projected average sales value is approximately **₹27 million**, while the projected average order volume is approximately **41,000 orders**. These figures represent forecast estimates based on historical patterns and may change as actual results develop ,results should be interpreted as directional estimates only.





The Business/Corporate segment shows a different projected pattern, with both sales and order volume declining. The forecast estimates an average of approximately **153 orders** and **₹108,000 in sales** for this segment. These figures are also predictive estimates and should therefore be interpreted as indicators of potential future trends rather than guaranteed outcomes.





Business Implications

- The analysis reveals a decline in order volume that has led to a drop in sales, indicating a clear correlation between the two. This trend warrants monitoring to determine whether the decline is temporary or persistent.
- Amazon generates higher sales rates and order volumes compared to the direct sales channel (merchant channel). Given the significant impact of the "Fulfillment by Amazon" (FBA) service on overall sales activity within the analyzed data, the service's performance, and the quality of service it delivers are critical aspects requiring ongoing monitoring.

- The "Set" category recorded the highest total sales, indicating a substantial contribution to overall revenue; consequently, monitoring demand for this category and tracking its performance is essential.
- Strong sales rates and high order volumes for "Medium"-sized products indicate that they are key contributors to overall sales activity, making demand for them a crucial factor to consider when evaluating product performance.
- The corporate/business sector accounts for only 0.79% of total sales; however, it records a higher average order value of 700.4 Rupees, compared to 648.1 Rupees for the retail/individual sector. This indicates that business sector orders have a higher average transaction value, despite their lower overall volume. However, forecasts also indicate a decline in the volume of business-sector orders, making this sector an area worthy of further analysis and monitoring.
- The high utilization rate of "expedited shipping" options for business orders suggests a preference within this sector for faster shipping services, although further analysis is required to understand the underlying reasons for this pattern.

Recommendations

- **Explore opportunities to increase sales from the Corporate/Business segment.** Although this segment currently represents a small share of total orders and sales, its average order value is higher than that of Retail/Customers. Further understanding this segment and increasing its order volume may provide an opportunity to increase sales.
- **Maintain the availability and performance of high-performing categories and sizes**, particularly the Set category and **Medium** size, which recorded the highest sales and order volume. Further analysis should be conducted to understand why these products perform better and which customer characteristics or purchasing patterns contribute to their demand.
- **Collect and analyze additional customer data in future analyses** to better understand customer characteristics, purchasing behavior, and the reasons behind differences in demand across products, sizes, and buyer types. This could support more targeted business decisions.
- **Conduct a deeper analysis of the decline in sales and orders over time.** The decline should be investigated by examining order status, unfulfilled orders, returns, fulfillment methods, and shipping patterns to determine which factors may be associated with the decrease in orders.
- **Give greater attention to Expedited shipping**, particularly within the Amazon and Corporate/Business segments, as these segments showed greater use of Expedited shipping compared with Standard shipping. Maintaining the availability and service quality of Expedited shipping in these segments may help meet their observed shipping needs.

- **Use the forecast results as an indicator for further investigation rather than as a confirmed outcome.** Since the forecast indicates a potential decline in the Corporate/Business segment and fluctuations in overall sales and orders, future analysis should compare actual results with the forecast and investigate the factors contributing to any continued decline.

Limitations

- The dataset covers a limited period, which restricts the ability to assess long-term sales and order trends.
- Some months in the dataset contain incomplete data, which may affect month-to-month comparisons.
- The dataset does not provide sufficient customer demographic information, limiting the ability to analyze customer characteristics and regional sales patterns.
- The analysis is based on the available fields in the dataset; therefore, some factors that may influence sales, such as marketing activities, pricing changes, customer preferences, and operational costs, could not be examined.
- The forecast results are based on historical patterns in the available data and should be interpreted as estimates rather than guaranteed future outcomes.

Conclusion

This analysis provided an overview of Amazon India sales performance by examining orders, sales, fulfillment methods, buyer types, product categories, sizes, and shipping methods. The analysis showed a decline in sales and order volume during the analyzed period, while Amazon recorded higher sales and received orders than Merchant. Retail/Customers accounted for the vast majority of sales and orders, where as Corporate/Business represented a much smaller share but recorded a higher average order value. At the product level, Set recorded the highest total sales, while Medium recorded the highest overall sales and order volume. The analysis also showed greater use of Expedited shipping within the Amazon and Corporate/Business segments. The forecast indicated potential fluctuations in overall sales and orders and a potential decline in the Corporate/Business segment. These results highlight areas that can be monitored and analyzed further as more data becomes available. Overall, the analysis demonstrates how sales, order, fulfillment, product, buyer, and shipping data can be combined to identify patterns and provide a basis for future business analysis and decision-making.